DATE OF CONSULT: 22-Nov-2023
PATIENT: MORRIS, Debra
MRN: AML063
DOB: 13-Jul-1964

University Cancer Institute - New Patient Consultation Note

REFERRING DIAGNOSIS: Newly Diagnosed Acute Myeloid Leukemia.

HISTORY OF PRESENT ILLNESS:
Ms. Morris is a 59-year-old female who presents for a new patient consultation to discuss her diagnosis and establish a plan of care. She has a history of well-controlled rheumatoid arthritis (on weekly methotrexate) and was feeling well until about one month ago. She initially developed what she believed was a viral illness with profound fatigue and body aches. When the fatigue persisted and she developed a large, non-traumatic hematoma on her thigh, she sought medical attention. Her labs revealed severe pancytopenia, and she was referred for an urgent bone marrow biopsy on November 20, 2023.

REVIEW OF OUTSIDE RECORDS:

- **Bone Marrow Biopsy Report (20-Nov-2023):**
    - Hypercellular marrow (90%) extensively replaced by myeloblasts (estimated 80% of cellularity).
    - Immunohistochemistry confirms myeloid lineage.
    - **Molecular NGS Panel:** Positive for a somatic mutation in **ASXL1**. No other significant mutations (FLT3, NPM1, IDH1/2, TP53 all negative).
    - **Cytogenetics:** Normal female karyotype (46,XX).

ASSESSMENT:
Ms. Morris is a 59-year-old female with de novo Acute Myeloid Leukemia. Based on the presence of a mutated ASXL1 gene, she is classified under the **ELN 2022 Adverse-risk** category, despite having a normal karyotype. This mutation confers a high risk of relapse with conventional chemotherapy alone.

DISCUSSION AND PLAN OF CARE:
I met with Ms. Morris and her husband for over an hour today. We had a detailed, comprehensive discussion regarding the diagnosis, prognosis, and treatment options.

1. Prognosis: I explained that the "adverse-risk" classification means that while remission is possible, standard chemotherapy has a lower chance of providing a long-term cure, and the risk of the leukemia returning is high.

2. Treatment Options: We discussed two primary upfront treatment strategies:
* **Option A: Intensive Induction Chemotherapy (e.g., "7+3").** This involves a 4-week hospital stay with significant side effects (hair loss, severe mucositis, high risk of infection). While she is fit enough to be a candidate, the high-risk nature of her ASXL1 mutation makes this a less attractive option, as the goal should be to achieve the deepest possible remission before considering further steps like a stem cell

transplant.
* **Option B: Less-Intensive Combination Therapy (Venetoclax + Azacitidine).** This regimen combines a BCL-2 inhibitor (Venetoclax, an oral pill) with a hypomethylating agent (Azacitidine, a subcutaneous injection or IV infusion). The VIALE-A study has shown this combination to be highly effective in newly diagnosed AML patients, including those with high-risk mutations. It produces high rates of remission and is generally better tolerated than intensive chemotherapy, often managed primarily in the outpatient setting after an initial brief admission.

3. Shared Decision Making: After reviewing the risks and benefits of each approach, Ms. Morris expressed a strong preference for the Venetoclax + Azacitidine regimen, valuing its efficacy and potentially better toxicity profile. I concur that this is the optimal choice for her specific disease biology.

4. Plan:

- **Initiate Treatment:** We will admit her to the hospital on **November 27, 2023,** to begin her first cycle.
  - **Azacitidine** will be given for 7 days.
  - **Venetoclax** will be started on Day 1 with a 3-day dose "ramp-up" (100mg -> 200mg -> 400mg) to mitigate the risk of Tumor Lysis Syndrome (TLS). She will take the 400mg dose daily thereafter.
- **TLS Prophylaxis:** She will receive aggressive IV hydration and Allopurinol. We will monitor her electrolytes, uric acid, and renal function every 6-8 hours for the first 48-72 hours.
- **Antimicrobial Prophylaxis:** She will be started on prophylactic antiviral, antifungal, and antibacterial medications, as this regimen can cause significant neutropenia.
- **Rheumatoid Arthritis:** Her methotrexate will be held indefinitely. We will manage any RA flares symptomatically.
- **Long-Term Plan:** The goal is to achieve a deep remission. Once remission is confirmed (typically with a bone marrow biopsy after Cycle 1 or 2), we will immediately refer her for a consultation with our Bone Marrow Transplant team to be evaluated for an allogeneic stem cell transplant, which represents her best chance for a long-term cure.

The patient and her husband verbalize understanding of this complex plan and are in agreement to proceed. All questions were answered.

Dr. Benjamin Carter, MD
Hematology-Oncology